
6.3 POTENTIAL FOR MISUSE AND DUAL-USE RISKS

The dataset’s potential for misuse, particularly in hiring and evaluation, is a significant "dual-use" risk. A worst-case scenario involves the data being used to unfairly judge job candidates, especially early-career researchers. Our stated goal is to increase transparency in the peer review process and empower researchers with a broader view of the academic landscape, not to create a performance ranking tool. To guard against misuse, we will take the following steps: 1) *Explicit Use Guidelines*: The Paper Copilot website will feature prominent guidelines strongly discouraging the use of its data for hiring, promotion, or other high-stakes evaluations of individuals. 2) *Focus on Aggregates*: The platform will prioritize the visualization of aggregated, large-scale trends over individual-level data. 3) *Community Dialogue*: We are committed to engaging in an ongoing dialogue with the AI/ML community to monitor how the data is used and to develop stronger safeguards as needed.

6.4 DATA INTEGRITY, INACCURACY, AND BIAS

The dataset is subject to several sources of potential inaccuracy and bias.

Sampling and Representation Bias. For closed-review venues, the reliance on community submissions introduces a self-selection bias. While this is currently the only feasible method for tracking review dynamics at scale without official data access, we recognize its limitations. To rigorously quantify this sampling bias, we plan to conduct a comparative analysis between our community-submitted data and the official ground-truth data from NeurIPS 2025 once it is released.

Data Inaccuracy. Inaccurate data, whether from parsing errors or malicious submissions, could harm researchers’ reputations. The LLM-based affiliation extraction has a non-zero error rate. We will mitigate this by: 1) Clearly labeling the source of all data (e.g., “Official API”, “Community-Submitted”). 2) Providing transparent metrics on the known error rates of our extraction models. 3) Implementing validation checks for community submissions to flag anomalous entries.

Demographic and Geographic Bias. The dataset could contain demographic and geographic biases, posing a special risk to researchers from certain groups (e.g., by geography, gender, or race). This bias can manifest in several ways. For closed-review conferences, the dataset’s reliance on voluntary community submissions may skew it toward certain regions or institutions that are more culturally inclined to share data. Furthermore, some research groups have internal policies that discourage sharing review data even when it is possible to opt-out, which could lead to the systematic underrepresentation of those groups in our dataset.

Such imbalances could lead to skewed analyses, where trends observed in overrepresented groups are incorrectly generalized to the entire research community, potentially disadvantaging those in underrepresented regions. To actively address this, we are committed to the following actions:

- **Transparency via Datasheets:** We will publish and maintain a datasheet that transparently documents the known geographic and institutional distributions in our dataset, highlighting potential skews so that researchers can interpret the data with proper context.
- **Regular Bias Audits:** We will perform regular audits to quantify these biases. Where possible, we will compare the distribution in our data against available ground-truth statistics from conference organizers to measure the extent of the disparity.
- **Platform-Level Context:** On our platform, any analyses derived from potentially biased data sources, such as community submissions, will be clearly marked with disclaimers that explain the data’s limitations and advise against overgeneralization.

7 CONCLUSION

Paper Copilot provides a unified, large-scale platform for tracking peer review dynamics and talent trajectories across major AI / ML conferences. It enables comparative analysis of score trends, rebuttal effectiveness, and institutional influence, with datasets already adopted by both academia and industry. Key limitations include partial reliance on voluntary data contributions for closed-review venues and the need to further improve the accuracy of talent trajectory reconstruction. Future work

will focus on expanding venue coverage and enhancing metadata quality for more comprehensive and precise analysis.

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